

Spatio-Temporal Prediction of County-Level NO₂ in the DMV Region Using Satellite Data and Machine Learning

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Abstract

Monitoring nitrogen dioxide (NO₂) pollution is critical for public health and environmental studies, but traditional ground-based methods are often limited in spatial coverage. Satellite platforms such as Sentinel-5P provide near-real-time NO₂ measurements through platforms like Google Earth Engine (GEE), but the data often contains gaps due to cloud cover and atmospheric interference. A predictive machine learning model is developed to estimate county-level NO₂ concentrations in the Washington D.C. Metropolitan Area (DMV). The model was trained on 2020 DMV data gathered from satellite-derived land use indices (NDBI, NDVI), land surface temperature (LST), population, and temporal variables. Results show high predictive accuracy in internal validation ($R^2 = 0.837$). The paper further investigates spatio-temporal generalizability; while performance declined, it remained positive for temporal generalization to 2019 DMV data ($R^2 = 0.507$) and spatial generalization to 2020 Pennsylvania data ($R^2 = 0.411$). The analysis also reveals distinct NO₂ seasonality and a notable decline during the initial COVID-19 lockdowns in 2020, followed by a gradual recovery. It is shown that, with proper training and data normalization, the machine learning model can predict and fill gaps in air quality data. This model could be used in urban planning in areas that lack sensor coverage.

Keywords

Earth and Environmental Science, Geosciences, Air Pollution, Machine Learning, Remote Sensing.

Introduction

Air pollution is one of the biggest environmental problems we face today, with major risks to global public health and ecosystems. According to the World Health Organization (WHO), it is a leading environmental risk factor. It contributes to over 4 million premature deaths annually, primarily due to diseases such as heart disease, lung cancer, and acute respiratory infections in children.¹ Among the various atmospheric pollutants, nitrogen dioxide (NO₂) is particularly harmful and widespread in urban areas. It is primarily generated from human activities, like burning of fossil fuels in internal combustion engines (e.g., vehicles, particularly diesel engines), industrial processes, and power generation. NO₂ also plays a crucial role in the formation of other dangerous secondary pollutants, including ground-level ozone (O₃) and fine particulate matter (PM_{2.5}), which may exacerbate respiratory and cardiovascular ailments.² This is why accurate and timely monitoring of NO₂ concentrations at fine spatial and temporal resolutions is important for effective public health protection planning and emission control policy.

Despite the critical need for complete air quality data, traditional ground-based monitoring networks offer precise and real-time measurements but have significant limitations. These networks are often unevenly deployed due to their sparse locations and high costs of setup and maintenance. A single regulatory monitor can cost up to \$50,000, not including the ongoing operation costs. Even for the less accurate and more affordable air quality sensors, they cost \$100 to \$2,500.³ This cost barrier is a primary reason for the extensive data gaps, making it difficult to capture the details of pollutant levels across different terrains and to determine how populations are exposed. This situation is especially severe in rapidly urbanizing areas, remote regions, or places with complex geography, like mountains. For example, many counties in

the United States do not have a single EPA air quality monitor.⁴ There is a call for affordable, scalable alternatives that can improve the current monitoring systems.

The rise of satellite remote sensing offers a practical solution to the limitations of ground-based monitoring. Instruments like the Tropospheric Monitoring Instrument (TROPOMI) on the European Space Agency's (ESA) Sentinel-5P satellite, launched in 2017, provide daily global measurements of tropospheric NO₂ column densities at resolutions as fine as 5.5 km by 3.5 km.⁵ This allows for large area coverage at a relatively low cost. Beyond direct pollutant measurements, satellite-derived proxies provide important environmental and human context. For instance, the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Built-up Index (NDBI) serve as indicators of urban development and vegetation cover, respectively. Land surface temperature (LST) shows how temperature-driven atmospheric conditions affect pollution dispersion. Meanwhile, gridded population datasets highlight the human aspect of exposure. The combination of these diverse datasets through cloud computing platforms like Google Earth Engine (GEE) has greatly changed environmental research. GEE's petabyte-scale data catalog and parallel processing abilities allow researchers to process and extract important environmental features globally at scale quickly. This helps overcome the computational challenges that usually come with large geospatial datasets.⁶

Parallel to this, machine learning (ML) has become a powerful tool in environmental modeling. Unlike traditional regression methods, ML algorithms, particularly ensemble models like Extreme Gradient Boosting (XGBoost), can model complex, nonlinear relationships and interactions among variables, making them well-suited for integrating multisource satellite data. There are many research papers showing the success of such methods in estimating air quality. For example, Di et al. (2019) applied random forest models to estimate daily PM_{2.5} across the continental U.S. with high accuracy using satellite, meteorological, and land use data.⁷ Similarly, Venter et al. (2020) analyzed satellite-derived NO₂ during the COVID-19 pandemic and observed a noticeable pollution reduction in major cities worldwide due to lockdowns.⁸

In addition to ML models, researchers have also explored statistical frameworks that account for spatial and temporal components in pollutant dynamics. Early studies employed linear regression and correlation analysis using aerosol optical depth (AOD) as a predictor for PM_{2.5}; however, the linear correlation approaches underfit because column AOD maps to surface PM_{2.5} in ways that depend on humidity and boundary layer height. Mixed-effect models, introduced by Kloog et al. (2011), addressed the temporal variability by incorporating random effects, allowing coefficients to vary over time.⁹ These models were adopted for PM_{2.5} prediction across different regions, including the Mid-Atlantic U.S., Mexico City, Israel, and China. Likewise, Geographically Weighted Regression (GWR) accounted for spatial nonstationarity, as demonstrated by Hu (2009), van Donkelaar et al. (2015), and others across the U.S. Adaptive and hybrid models that blend spatial and temporal adjustments have shown improved accuracy over fixed-coefficient models. These advances underscore a key insight: spatial and temporal variation in air quality drivers must be explicitly modeled as a dynamic process, not a static snapshot.

While traditional statistical models offer clear interpretability, they often lack the flexibility to capture complex nonlinear relationships and interactions among high-dimensional environmental variables. In contrast, machine learning approaches such as XGBoost are well-suited for such tasks, enabling more nuanced and accurate predictive modeling. Despite their increasing adoption, many ML-based air quality models are trained on aggregated national datasets and are seldom evaluated for spatio-temporal generalizability. This represents a significant limitation, particularly for policy and transfer learning contexts where models developed in data-rich regions are expected to inform decisions in areas with limited ground-based monitoring.¹⁰

The predictive modeling remains essential to overcome data gaps, ensure temporal continuity, and enable inference in under-observed regions, as Google Earth Engine (GEE) provides satellite-based NO₂ observations. Data quality and preprocessing are important in training the model. For example, issues like cloud cover, retrieval errors, and coarse resolution frequently lead to incomplete satellite observations.

Even with standard GEE practices such as cloud masking to filter out corrupted pixels, significant gaps remain, particularly in rural or low-emission areas. Furthermore, many policy applications require consistent, gap-free data streams and the ability to make projections beyond current observations. Our spatial and temporal validation, performed at a proper scale to match the desired resolution, ensures the soundness of our predictive model for these real-world, policy-related applications across diverse settings.

While platforms like Google Earth Engine (GEE) provide accessible satellite-based NO₂ observations, the data quality and preprocessing are of critical importance for training an effective model. Cloud cover, retrieval errors, and coarse resolution can lead to incomplete satellite observations, particularly in rural or low-emission areas. Even with standard GEE practices such as cloud masking to filter out corrupted pixels, significant gaps remain. This is why predictive modeling remains essential: it allows us to overcome these data gaps, ensure temporal continuity, and enable inference in underobserved regions. Furthermore, since many policy applications require consistent, gap-free data streams and the ability to make projections beyond current observations, our spatial and temporal validation, performed at a proper scale to match the desired resolution, ensures the soundness of our predictive model for these real-world, policy-related applications across diverse settings.

This study aims to develop and evaluate a predictive model for county-level NO₂ concentrations in the Washington D.C. Metropolitan Area (DMV), which includes part of Virginia, Maryland, and close to the District of Columbia. The DMV is characterized by a diverse mix of urban, suburban, and rural counties, providing a good testbed for regional-scale environmental modeling. Our research leverages GEE for large-scale satellite data processing and XGBoost for predictive modeling. This study aims to:

- Predict monthly county-level NO₂ concentrations using 2020 satellite-derived variables, including NDVI, NDBI, LST, population density, and month.
- Evaluate temporal generalization by testing the model's performance on 2019 DMV data to assess year-over-year consistency.
- Evaluate spatial generalization by applying the 2020 DMV-trained model to Pennsylvania counties in the same year.
- Analyze NO₂ seasonal trends, especially focusing on the impact of the COVID-19 lockdowns and subsequent economic reopening on pollutant concentrations.
- Evaluate the broader benefits and challenges of using GEE and ML for air quality monitoring, including issues of satellite data availability, spatial resolution, and computational limitations.

By combining GEE and XGBoost, this research offers a scalable and reproducible framework for monitoring regional NO₂ levels. It also shows how cloud-based geospatial platforms can support data-driven public health and environmental decisions at the sub-national level.

Methods

This section details the methodologies employed for data acquisition, preprocessing, model development, and validation strategies to predict county-level NO₂ concentrations.

2.1. Study Area

The primary study area for model training and internal validation was the Washington D.C. Metropolitan Area (DMV), including parts of Virginia and Maryland. For assessing spatial generalizability, the state of Pennsylvania was selected as an external, geographically distinct region. While bordering the DMV study area, Pennsylvania has a different geography and climate. It has more varied topography, including mountainous regions, compared to the more coastal-influenced DMV region.

2.2. Data Sources and Preprocessing

All satellite imagery and geospatial datasets were processed on the Google Earth Engine (GEE) cloud-based platform. The GEE's powerful parallel processing capabilities enabled the efficient handling of large environmental datasets. GEE's vast data catalog, coupled with its free access for academic and research projects, made it an ideal tool for the study. The platform's JavaScript API was used to perform all geospatial operations.

The following data products were utilized:

- **Nitrogen Dioxide (NO₂):** Tropospheric NO₂ column number density data were obtained from the Copernicus Sentinel-5P (S5P) Level 3 dataset (COPERNICUS/S5P/OFFL/L3_NO₂). Given that reliable S5P data has been available since late 2017, I selected 2019 and 2020 as the study years to ensure temporal consistency for analysis and validation.
- **Population Density:** Global population density data at 100-meter resolution was sourced from the WorldPop dataset (WorldPop/GP/100m/pop) to represent human activity.
- **Vegetation Indices (NDBI, NDVI):** The Normalized Difference Built-up Index (NDBI) and Normalized Difference Vegetation Index (NDVI) were derived from Copernicus Sentinel-2 (COPERNICUS/S2_HARMONIZED) imagery. These indices serve as proxies for urban development and green spaces, respectively. I calculated NDBI using the shortwave infrared (Band 11) and near-infrared (Band 8) bands, and NDVI using the near-infrared (Band 8) and red (Band 4) bands. Cloud masking was applied to Sentinel-2 imagery to ensure data quality.

$$NDBI = \frac{SWIR - NIR}{SWIR + NIR} \quad NDVI = \frac{NIR - RED}{NIR + RED}$$

Where:

SWIR = Shortwave Infrared reflectance (Band 11)

NIR = Near Infrared reflectance (Band 8)

RED = Red reflectance (Band 4)

- **Land Surface Temperature (LST):** Daytime Land Surface Temperature (LST_Day) was extracted from the MODIS Terra Land Surface Temperature and Emissivity 8-Day Global 1km data product (MODIS/006/MOD11A2). This covers a key meteorological factor affecting the dispersion of NO₂. The LST values were scaled by a factor of 0.02 as per the dataset's specifications.

For each feature, monthly mean images were generated. These monthly images were then spatially aggregated by reducing the mean value within each county boundary, defined by the FAO Global Administrative Unit Layers (GAUL) 2015 dataset (FAO/GAUL/2015/level2). This process resulted in a dataset of monthly, county-level feature values for all variables. To address the skewed distribution of NO₂ density and improve model performance, a natural logarithm transformation was applied to the NO₂ target variable. The month (1-12) was included as a temporal feature to account for seasonality.

2.3. Machine Learning Model

Extreme Gradient Boosting (XGBoost) regression model is widely used in the machine learning field for its high accuracy, efficiency, and scalability. This study chose XGBoost due to its ability to handle missing values well, a common issue in environmental datasets. Most importantly, its superior performance in

capturing complex, non-linear interactions among environmental features is a key requirement for accurately modeling NO₂ dynamics.

Given the DMV dataset's modest size relative to the number of predictors, overfitting is a major risk. XGBoost was used as it controls model complexity through shrinkage (learning rate), shallow trees (max_depth), and L1/L2 regularization (row/column subsampling). Trees are added sequentially to fit residuals, but training is capped with early stopping on a spatially blocked validation fold, limiting memorization of local noise. XGBoost also routes missing values via learned default splits, avoiding ad-hoc imputation that can further amplify overfitting.

XGBoost also provides feature importance diagnostics that help interpret the model rather than treat it as a black box. It reports gain-based importances and cross-checks them with Shapley Additive Explanations (SHAP) values to see which predictors: population density, NDVI, NDBI, LST, and seasonality, most influence NO₂ estimates. Because some inputs are correlated, these importances are treated as associative, not causal.

The model was trained on preprocessed monthly, county-level DMV data for 2020. Environmental pollution data, such as NO₂ concentrations, often show positive skew: most values are moderate, but a small number of very high observations stretch the upper tail. To reduce skew and stabilize variance, this study applies a log_p transform, $\log(1 + x)$, to handle zeros safely and near-zero values. This preprocessing is especially helpful when training regression models like XGBoost because:

- **Improved Model Stability:** Log transformation reduces heteroscedasticity (non-constant variance), so residuals are more even across concentration ranges.
- **Better Handling of Outliers:** Extreme NO₂ values are downweighted, preventing the model from overfitting to rare, high-pollution events.
- **Enhanced Generalization:** By smoothing the data distribution, the model can better capture overall patterns without being biased toward the majority of low-concentration samples.

After training on log-transformed NO₂ values, the predictions are back-transformed using the exponential function to interpret results in the original measurement units. This preserves interpretability in physical units while retaining the stability benefits of learning in log space. This study tuned hyperparameters with grid search and k-fold cross-validation with early stopping to curb overfitting. The search varied learning_rate (η), max_depth, subsample, n_estimators, and L1/L2 regularization (reg_alpha, reg_lambda), balancing accuracy with model complexity.

2.4. Validation Strategy

To thoroughly assess the model's performance and its ability to generalize, I used a multi-step validation strategy:

- **Internal Validation:** The model's performance was first evaluated on unseen data from within the 2020 DMV dataset using cross-validation. RMSE (Root Mean Squared Error; higher values indicate a better model fit) and R² were monitored for early stopping to prevent overfitting. This evaluates the model's ability to generalize to new, but similar, observations from the same spatial and temporal context, ensuring the model did not overfit to the training data.
- **Temporal Generalization:** To test the model's ability to predict NO₂ concentrations in a different time period, the model trained on 2020 DMV data (with hyperparameters frozen) was applied to the 2019 DMV dataset. It shows whether a model fit in 2020 still works in 2019. Any drop points to year-to-year shifts in emissions, weather, or activity that fixed predictors miss.

- **Spatial Generalization:** To evaluate the model's transferability to an entirely new geographical area, the 2020 DMV-trained model was applied (without retraining) to the 2020 Pennsylvania dataset. Applying the DMV-trained model to Pennsylvania checks whether it transfers across geography, different density, land use, and industry, as it would in places with few monitors.

2.5. Evaluation Metrics

The performance of the predictive model across all validation scenarios was quantified with two standard metrics:

- **Coefficient of Determination (R^2):** This metric indicates the proportion of the variance in the dependent variable (NO_2 density) that is predictable from the independent variables. Values closer to 1 indicate a tighter fit; values near 0 (or negative) indicate little explanatory power. Note that R^2 depends on the variability of the evaluation set and does not convey error magnitude.
- **Root Mean Squared Error (RMSE):** RMSE measures the average magnitude of the errors between predicted and actual NO_2 values. It provides an interpretable measure of predictive accuracy in the same units as the target variable ($\text{mol/m}^2 \times 10^6$), reflecting the typical deviation from the true NO_2 levels. Lower RMSE indicates better accuracy, and its squaring step makes it more sensitive to large errors, which is useful for identifying poor local performance.

R^2 and RMSE metrics summarize the variance and typical error magnitude, enabling consistent comparison across the internal, temporal, and spatial validation scenarios.

Results and Discussion

The XGBoost model demonstrated predictive performance in estimating monthly county-level NO_2 concentrations at resolution within the DMV region for 2020. The internal validation, performed via cross-validation on held-out data, achieved an R^2 of 0.837 and an RMSE of $4.31 \text{ mol/m}^2 \times 10^{-6}$. These results suggest that the model can capture the underlying relationships between environmental covariates and NO_2 pollution at the county level. The scatter plot of actual versus predicted NO_2 values (Figure 1) reveals a tight clustering of points along the 1:1 line, confirming that the model's predictions closely match observed values in the DMV training domain. This strong fit supports the use of machine learning to bridge the gaps left by sparse ground-based monitoring systems.

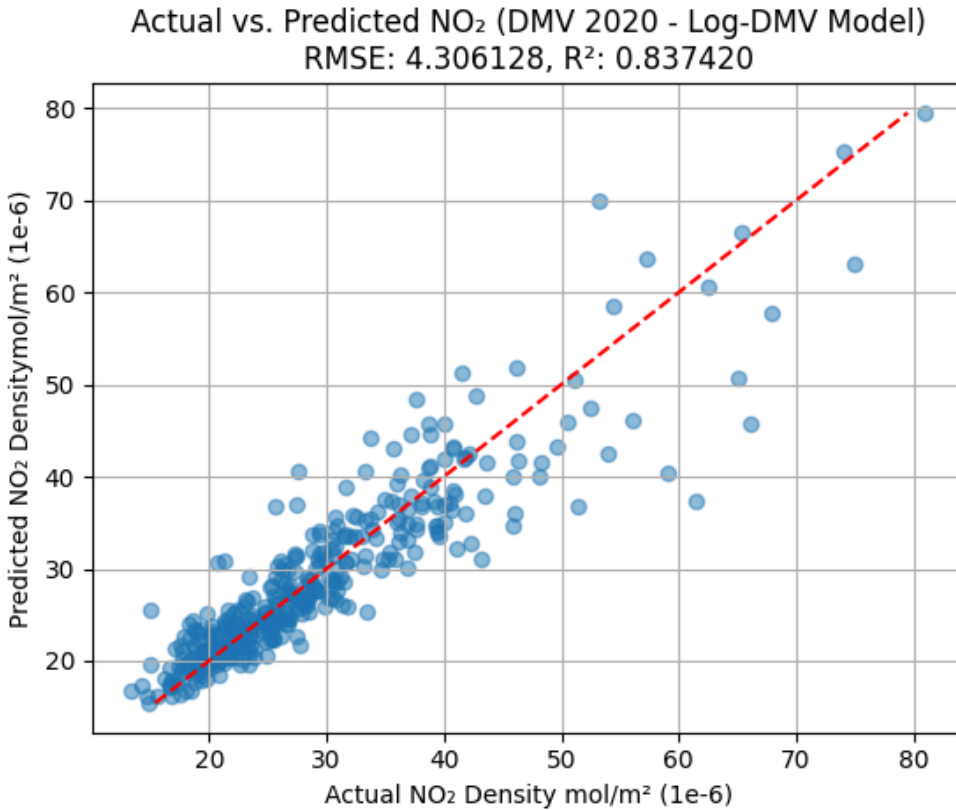


Figure 1. Actual vs. Predicted NO₂ Density (Log-scale Trained Model). The scatter plot shows a tight clustering of points along the 1:1 line, indicating that the model's predictions closely match the observed values. This confirms the model's high predictive accuracy.

An analysis of feature importance from the XGBoost model showed that population density was the strongest predictor (F-score: 1076.0). This finding reflects the close link between human activities, like traffic, energy use, and urban infrastructure. NDBI (F-score: 957.0) further highlighted the impact of urbanization. Built-up areas usually have higher emissions due to increased vehicle use, industrial activities, and less vegetation.

Land Surface Temperature (LST, F-score: 777.0) also played an important role. Higher temperatures affect how pollutants spread. NDVI (F-score: 727.0) showed an inverse relationship with NO₂. This aligns with the idea that vegetation helps improve air quality by filtering out pollutants.

The month variable (F-score: 600.0) captured seasonal changes in NO₂ levels, influenced by factors like shifts in weather, photochemistry, and human behaviors (for example, winter heating and summer travel).

These results on feature importance (Figure 2) clarify the complex interactions between urbanization, environmental conditions, and seasonal cycles in shaping NO₂ levels. They point to both where emissions come from and how the surrounding environment shapes NO₂.

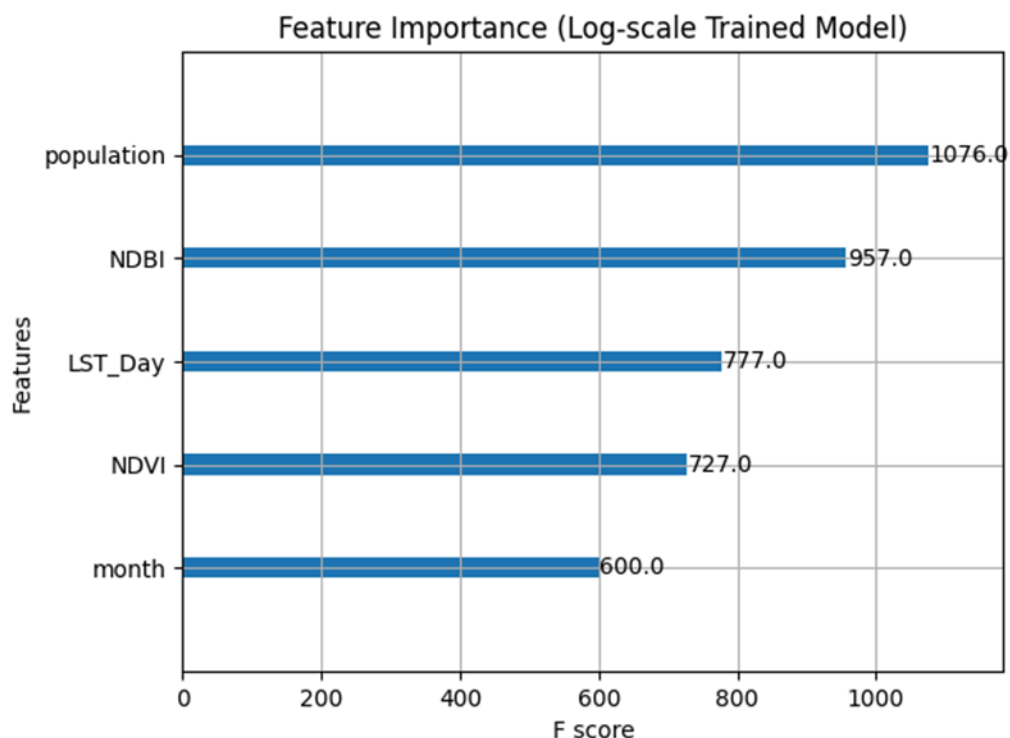


Figure 2. Feature Importance (Log-scale Trained Model). The bar chart displays the F-scores for each feature used in the XGBoost model, with a higher score indicating greater importance. Population density was the most important predictor, followed by NDBI, LST_Day, NDVI, and month, which reflects the relationship between human activities, urbanization, and environmental factors on NO₂ levels. The F-score indicates statistical association, not causation.

The correlation matrix of the features and NO₂ supports these findings. The population had a positive correlation with NO₂ at 0.48, and NDBI was also positive at 0.46. This indicates that more populated and built-up areas tend to have higher NO₂ levels. In contrast, NDVI had a strong negative correlation with NO₂ at -0.6. This suggests that areas with denser vegetation have lower NO₂ concentrations. Daytime land-surface temperature (LST_Day) shows a moderate negative correlation with NO₂ of -0.50, consistent with warmer days promoting boundary layer mixing that dilutes near-surface NO₂. The month variable had a weak positive correlation at 0.069. These correlations reveal the complex interaction of environmental and human factors on air quality (Figure 3).

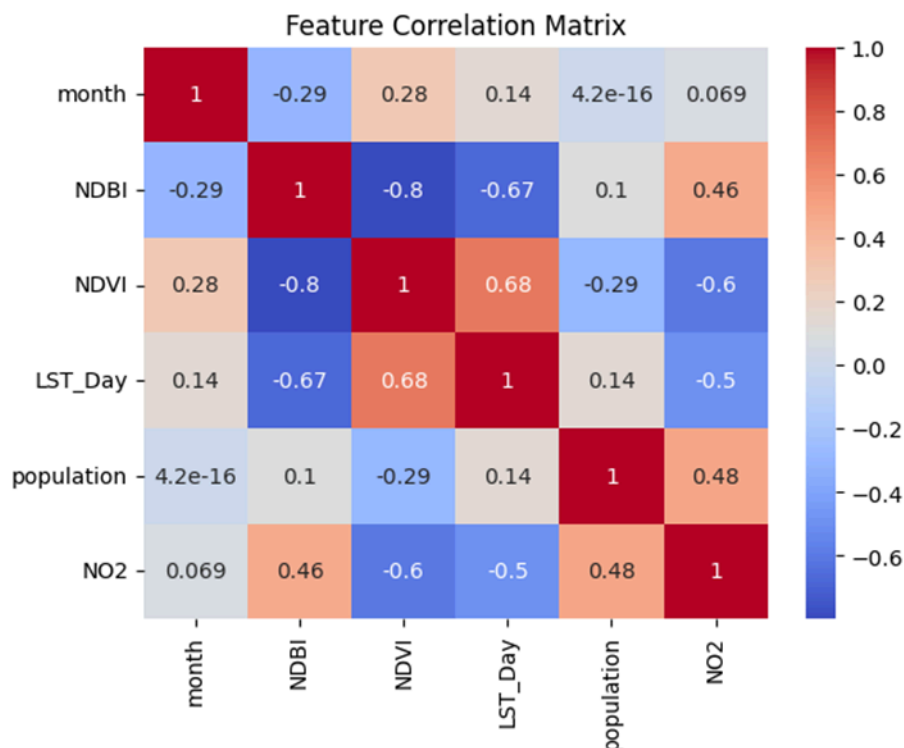


Figure 3. Feature Correlation Matrix. The matrix shows the correlation coefficients between the different features and NO₂. Population and NDBI show a positive correlation with NO₂, indicating higher pollution in more populated and built-up areas. NDVI and LST_Day have a negative correlation with NO₂, suggesting that denser vegetation and higher temperatures are associated with lower pollution levels.

The model's temporal generalization was assessed by applying the 2020 DMV-trained model to 2019 DMV data. This yielded an R^2 of 0.5074 and an RMSE of $8.89 \text{ mol/m}^2 \times 10^{-6}$ (Figure 4). While the R^2 is lower than the internal validation ($R^2 = 0.837$), the model still demonstrates a reasonable ability to capture year-to-year NO₂ variability within the same region. This result suggests that key predictors, such as land use, temperature, and population, retain consistent relationships with NO₂ across different time periods. It holds even when unmodeled factors such as policy changes, weather anomalies, or emission shifts occur.

The model tends to underpredict at higher NO₂ concentrations, as shown by the widening residuals in the upper range of the scatter plot. This pattern may reflect real-world changes between 2019 and 2020, including unaccounted environmental factors. The shrinkage toward the training range when back-transforming from the log scale may also contribute to the underprediction at higher NO₂ levels. Nevertheless, the temporal transferability is promising, supporting the model's potential use for monitoring pollutant trends over time.

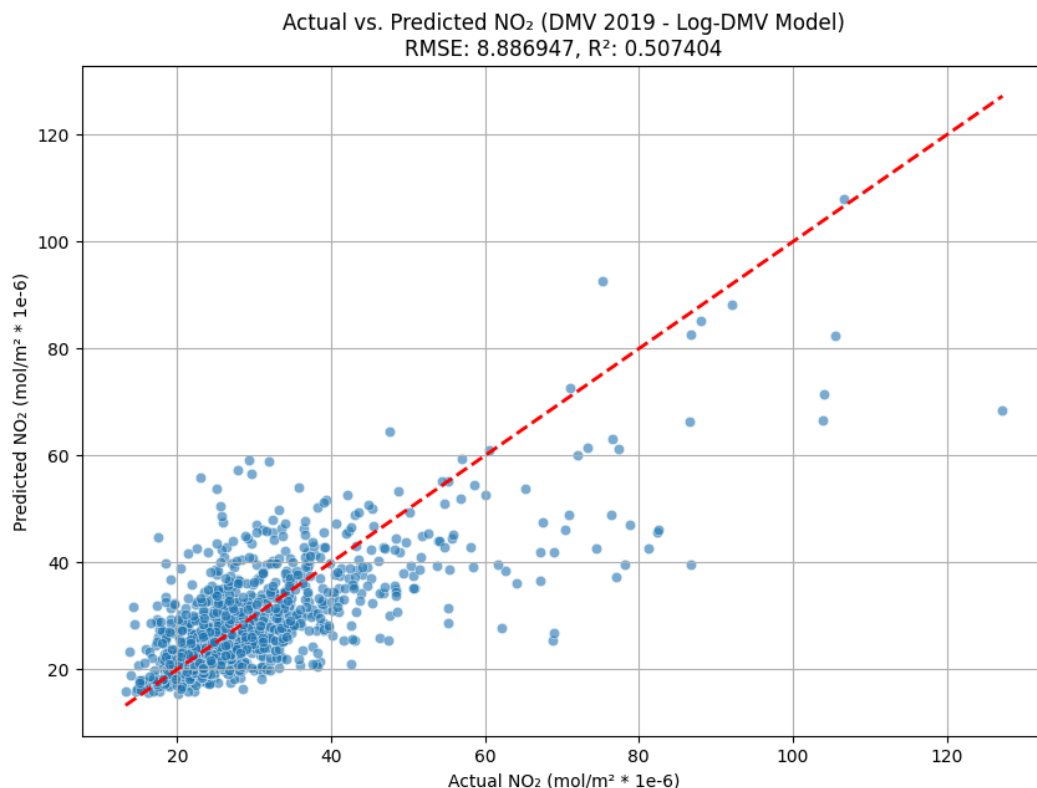


Figure 4. Actual vs. Predicted NO₂ (DMV 2019 - Log-DMV Model). This scatter plot shows the model's performance when applied to 2019 data, demonstrating its temporal generalization. The model achieved a reasonable R² of 0.507 and an RMSE of 8.89 mol/m² × 10⁻⁶, indicating its ability to capture year-to-year variability. The plot also shows that the model tends to underpredict at higher NO₂ concentrations, which may be due to unmodeled environmental factors or the log scale back-transformation.

For spatial generalization, the model's performance on 2020 Pennsylvania data resulted in an R² of 0.411 and an RMSE of 11.53 mol/m² × 10⁻⁶ (Figure 5). Accuracy is lower than in the internal DMV and 2019 temporal tests, reflecting domain shift across regions. The reduced R² highlights the challenges of transferring machine learning models across geographically distinct regions. Differences in urban form, emissions mix, traffic intensity, and industrial activity, plus Pennsylvania's larger share of rural/semi-rural counties, are only partly captured by a model tuned on DMV features.

Despite these limitations, the positive R² indicates the model still captures over half of NO₂ variability across space, providing some level of generalizability. This is particularly valuable for regions with limited ground-based air quality monitoring, where satellite-driven models can offer preliminary estimates to guide environmental assessments.

Future improvements could include adding region-specific features, such as source inventories, road density, and topography, incorporating additional emissions datasets, or applying transfer learning techniques to better adapt models to new spatial domains.

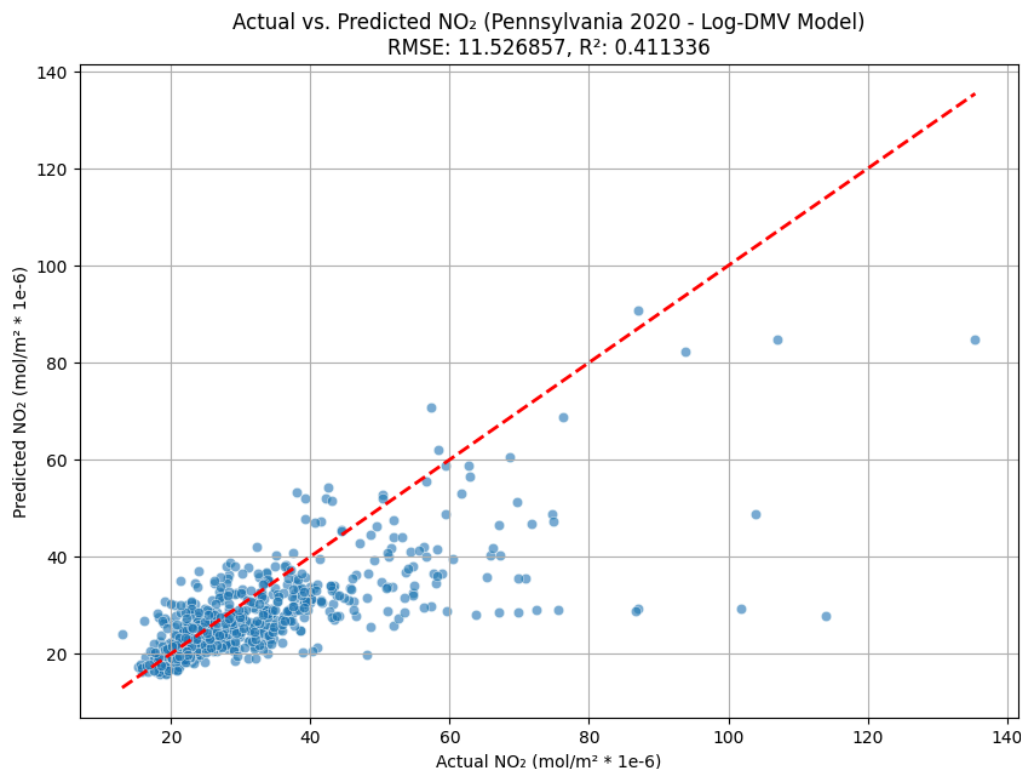


Figure 5. Actual vs. Predicted NO₂ (Pennsylvania 2020 - Log-DMV Model). This scatter plot displays the model's performance when applied to data from a new geographical region, Pennsylvania. The lower R² of 0.411 and higher RMSE of 11.53 mol/m² × 10⁻⁶ compared to the DMV-specific tests reflect the challenges of transferring machine learning models across different regions. Despite the reduced accuracy, the positive R² suggests the model still captures some of the NO₂ variability in a new domain, which can be valuable for regions with limited ground monitoring.

Environmental pollution data, such as NO₂ concentrations, often show positive skew: most values are moderate, but a small number of very high observations stretch the upper tail. This heavy-tailed pattern is visible in the left panel of Figure 6.

Applying a logarithmic transform compresses large values while preserving rank. We use the log_p transform, $\log(1 + x)$, to handle zeros safely and near-zero values. As shown in the right panel, the transformed distribution is closer to symmetric (approximately normal), with more uniform variance and less influence from extreme values. This preprocessing is especially helpful when training regression models like XGBoost because:

- **Improved Model Stability:** Log transformation reduces heteroscedasticity (non-constant variance), so residuals are more even across concentration ranges.
- **Better Handling of Outliers:** Extreme NO₂ values are downweighted, preventing the model from overfitting to rare, high-pollution events.
- **Enhanced Generalization:** By smoothing the data distribution, the model can better capture overall patterns without being biased toward the majority of low-concentration samples.

After training on log-transformed NO_2 values, the predictions are back-transformed using the exponential function to interpret results in the original measurement units. This preserves interpretability in physical units while retaining the stability benefits of learning in log space.

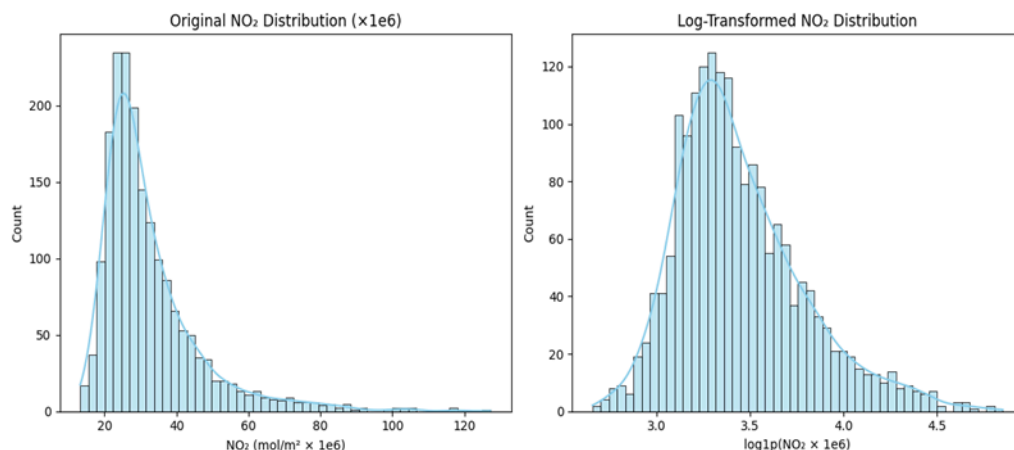


Figure 6. Distribution of NO_2 concentrations in the DMV region (2019). The left panel displays the original NO_2 concentration data, which has a positively skewed distribution with a heavy tail. The right panel shows the data after a $\log_{1p}$ transformation, which makes the distribution more symmetric and approximately normal. This transformation improves model performance by reducing the influence of outliers and making the variance more uniform.

The left panel shows the original NO_2 data ($\text{mol/m}^2 \times 10^6$), while the right panel displays the $\log_{1p}$ -transformed distribution. Log transformation reduces skewness, normalizes the data, and improves model performance by stabilizing variance and reducing the impact of outliers.

Analysis of the monthly NO_2 pollution trend in the DMV region from 2018 to 2023 shows clear seasonal cycles and the effects of the COVID-19 pandemic. Generally, NO_2 concentrations are higher in winter because of increased heating and more frequent inversions that trap pollutants near the ground. In contrast, summer months have lower NO_2 levels due to quicker breakdown of NO_2 , higher mixing heights, and less heating demand.

The plot in Figure 7 shows a sharp drop in NO_2 levels starting in March 2020, which coincides with the beginning of COVID-19 lockdowns. This decline continued through the first reopening phase in July 2020, followed by a gradual increase in NO_2 during 2021 as economic activities picked up again. The timeline matches policy changes, including phased reopening and shifts in mobility patterns. Keep in mind that the plot indicates association, not direct causation.

From 2021 to 2023, NO_2 levels returned to typical seasonal cycles but at lower peaks compared to years before the pandemic. This might reflect long-term changes in emissions, such as more remote work and changes in transportation habits. On a typical 2022 workday, roughly 44% of workers were teleworking, implying fewer in-person commutes than pre-pandemic, according to the statistics. Monitoring is needed to determine whether this trend continues.

These findings show the model's ability to capture real-world temporal variability in NO_2 driven by both seasonal factors and large societal events.

Monthly NO₂ Pollution Trend (DMV Region, 2018-2023) with COVID-19 Events

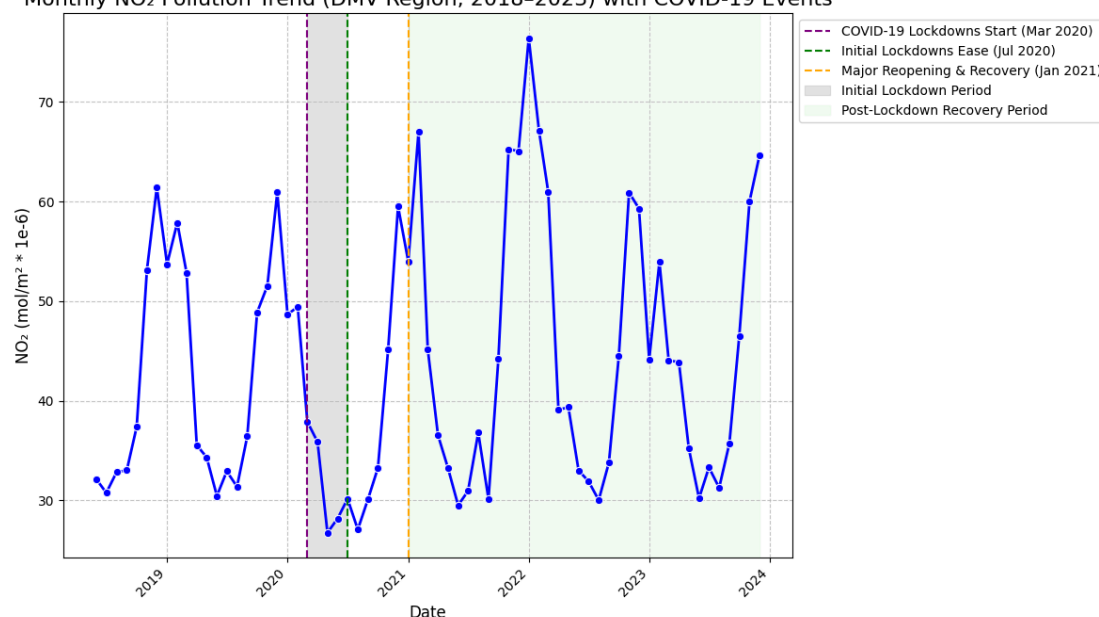


Figure 7. Monthly NO₂ Pollution Trend (DMV Region, 2018-2023) with COVID-19 Events. This plot illustrates the monthly changes in NO₂ concentrations in the DMV region, revealing clear seasonal cycles with higher levels in winter and lower levels in summer. It also highlights a sharp drop in pollution that began with the initial COVID-19 lockdowns in March 2020, followed by a gradual recovery in 2021 as economic activities resumed. The data suggest that post-pandemic NO₂ peaks remained at a lower level than pre-pandemic years, possibly indicating long-term changes in human behavior, such as increased remote work.

Although predictive performance was strong within the training domain, reduced accuracy on external data highlights the need for region-specific calibration. Factors like unobserved emissions sources, industrial zoning, or meteorological variation (e.g., boundary-layer depth and wind regimes) may introduce bias. Additionally, satellite-derived data may be limited by cloud cover and temporal gaps, particularly during winter months.

The interactions of gradient-boosted tree splits constrain interpretability. To make it more transparent, future work could pair the model with explainability tools (e.g., SHAP, accumulated local effects) and ingest higher resolution emissions data (point sources, stack heights) to sharpen spatial gradients.

While the model performs well in the 2020 DMV dataset, transferring it to new regions or years benefits from lightweight adaptation: simple scale recalibration, adding region-specific predictors, or limited fine-tuning on a small local label set. These steps help mitigate shifts in emissions mix, zoning patterns, and local weather that may degrade accuracy.

Satellite-derived data, while powerful, have inherent limitations. Cloud cover, snow, and retrieval screening may reduce data quality, especially during winter months. Satellites measure the total amount of a pollutant in a vertical column of the atmosphere, known as tropospheric column density. This isn't the same as the surface concentration that people actually breathe. Thus, vertical mixing, boundary-layer height, and retrieval assumptions can decouple column NO₂ from ground-level conditions in some situations.

Another limitation lies in model interpretability. Gradient-boosted trees like XGBoost often outperform linear models but are less transparent. Understanding complex feature interactions is hard to trace, suggesting the need for explainable AI tools and higher resolution inputs (e.g., emissions and traffic detail) to improve model transparency and spatial precision.

Conclusion

This study developed and validated an XGBoost model to predict county-level NO₂ in the Washington, DC metropolitan area (DMV) using satellite-derived inputs processed in Google Earth Engine (GEE). By integrating environmental and human indicators: Normalized Difference Built-up Index (NDBI), Normalized Difference Vegetation Index (NDVI), land-surface temperature (LST_Day), population density, and seasonality (month), the model captured major patterns in atmospheric NO₂ variability.

Within the training domain (2020 DMV), cross-validated performance was strong ($R^2 = 0.837$; RMSE = $4.306 \text{ mol/m}^2 \times 10^{-6}$), indicating that these predictors explain a substantial share of county-level variation. This level of accuracy is promising for regional air quality assessments in areas lacking dense ground-based monitoring.

Feature importance summaries ranked population density highest, consistent with the influence of transportation and energy use on NO₂. Built-up cover (NDBI), land-surface temperature, and vegetation (NDVI) also ranked highly, underscoring links between urbanization, meteorology, and air quality. It should be noted that feature importance ranking shows statistical association, but not causal mechanisms.

Importantly, this study tested the spatio-temporal generalizability of the model, an area not always reported in environmental machine learning research. The model showed reasonable temporal transfer to 2019 DMV data ($R^2 = 0.507$), indicating that learned patterns carry across years. Spatial transfer to 2020 Pennsylvania was lower ($R^2 = 0.411$), reflecting the difficulty of moving models across regions with different emission mixes, urban form, and meteorology. Even so, the positive out-of-domain R^2 values suggest scope for broader regional use with region-specific calibration and modest adaptation.

The analysis of NO₂ trends from 2018 to 2023 captured clear seasonal cycles and a pandemic-era signal: a sharp decline beginning in early 2020, followed by a gradual rebound during reopening in 2021. Peaks did not fully return to many pre-pandemic levels through 2023, consistent with potential longer-run changes in travel and activity patterns, remote work, cleaner fleets, and yearly meteorology. It is interpreted as associations, not causal estimates.

This work shows that integrating satellite remote sensing, cloud platforms (GEE), and machine learning can deliver scalable, near-real-time regional monitoring. The approach is cost-effective, reproducible, and it is particularly useful where ground monitors are sparse. It provides actionable inputs for public health, planning, and policy, such as screening maps and trend indicators. There is also room for improvement via region-specific features, emissions data, and hyperparameter tuning.

Looking ahead, this work sets up several extensions: integrate higher resolution emissions data (e.g., point sources, stack parameters), traffic flow/speed models (road density), and real-time meteorological fields (boundary-layer height, wind, stability) to sharpen spatial predictions. The model could also be paired with explainable AI (SHAP, accumulated local effects) to clarify drivers for policymakers and added with uncertainty quantification so maps include confidence information. Scaling to multistate or national datasets and testing temporal sequence models (e.g., state-space or LSTM) could capture persistence and lagged effects and further improve forecasts for management use.

In conclusion, the study presents a machine learning-based framework for regional NO₂ prediction that is scalable and reproducible using cloud resources. It offers a practical tool for air quality assessment and decision support in places with sparse monitors, while leaving room for targeted enhancements (region-specific features, emissions data, explainability, and uncertainty estimates) as the system is deployed more broadly.

Future Research Directions

- Integrate emissions and traffic data (e.g., inventories, road density) to represent sources more directly.
- Applying temporal models (e.g., LSTM) to better capture long-term pollution dynamics.
- Using explainable AI methods (e.g., SHAP or LIME) to clarify how each feature contributes to predictions.
- Implementing probabilistic models to quantify uncertainty and provide confidence intervals for predictions.

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